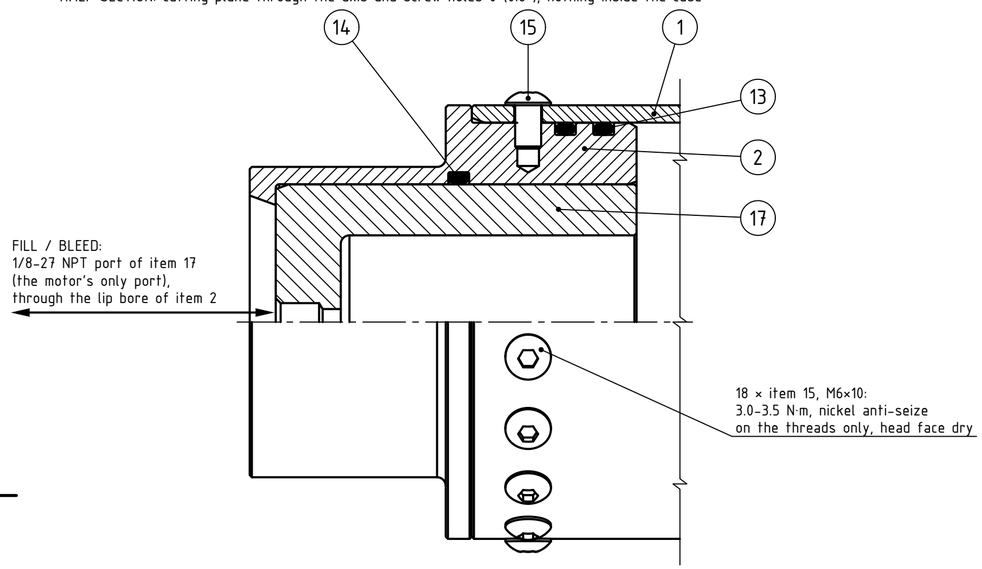
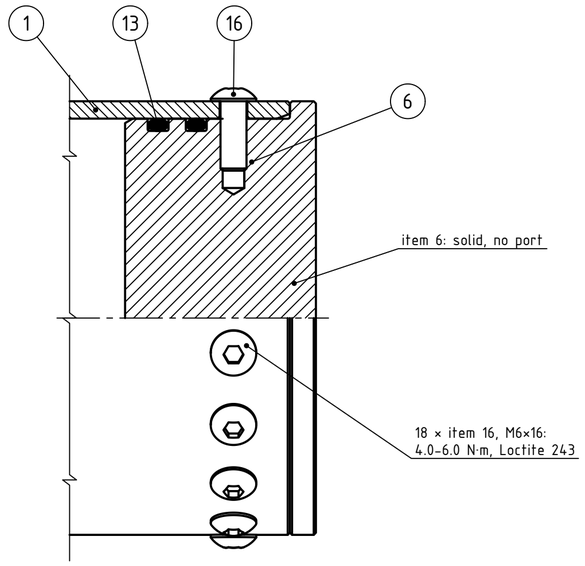


B HALF SECTION: cutting plane through the axis and screw holes 0 (0.8°); nothing inside the case C



B (1:1)



C (1:1)

PROOF BUILD

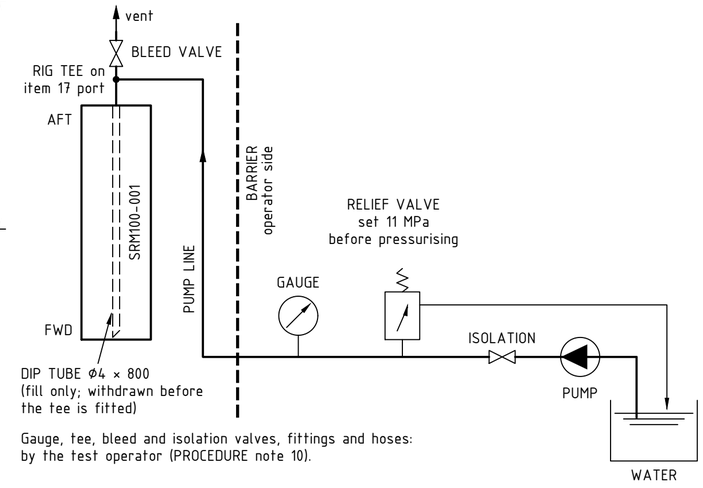
1. Build list: the flight case item 1, both closures (items 2, 6), all five O-rings (4 × item 13, item 14) and all 36 screws (items 15, 16), with proof dummy item 17 in the insert's gland. No graphite insert item 3 (it must stay dry; a cracked one is scrap), no insulators, liner, grain or cap: nothing inside the case.
2. Each closure goes to its own case end: match-drilled, serialised pair. θ0: on the case, hole 0, the clocking line scribed across the pair and the AFT / FWD marking; on each closure, puller P1 at θ 10° (item 2 r 41, item 6 r 37.5). Hole 0 is at 0.8°, 9.2° clockwise of P1 seen from forward (counter-clockwise seen from aft).
3. O-rings items 13, 14: FKM 90A, Parker V0709-90 or equal. Lube: Parker Super O-Lube or DC-55, thin film. Fit item 14 in item 2's bore groove, then 2 × item 13 in each closure's land grooves. Items 2 and 6 are cut to their own case end's bore A (fits table).
4. Slide item 17 into item 2 by hand, aft end first, until it seats on the lip. Never press or knock it (view B). Check: its forward face lies flush with item 2's forward face to 0.10 below it.
5. Hole plugs (tooling): before each closure goes in, push 18 flat-ended acetal plugs, ϕ6.35 × 3.85 with an outside head, into that end's holes; their ends sit about 0.15 mm shy of the bore, so no O-ring is dragged across an open hole edge. Pull each plug as its screw goes in.
6. Enter each closure at its own end (item 2 AFT, item 6 FWD) with its hole 0 on the case's hole 0 and clocking line, until its flange butts on the case end face: zero gap.
7. Torques as for flight. Aft: 18 × item 15 into item 2 (304), 3.0–3.5 N·m, nickel anti-seize on the threads only, head face dry. Forward: 18 × item 16 into item 6 (6061), 4.0–6.0 N·m, Loctite 243. Criss-cross, in the boxed order of SRM100-000 sheet 2 (aft) / sheet 3 (fwd), 3 passes: about half the minimum torque (1.5 / 2.0 N·m), then to torque, then a check pass hole by hole round the circle. Do not mix the lengths: an M6×16 in the holder bottoms with its head proud; an M6×10 in the forward closure engages only 1 d.
8. Before the first proof, record the hole baseline of all 36 case holes (SRM100-101 record R7), keyed to the serialised pair: the axial length of each hole, and the ligament from the case end face to the hole edge (nominal 13.0 − 3.2 = 9.80). Each hole is judged against its own baseline only, never an absolute ligament limit (SRM100-101 R7).
9. Mark the case OD stations S1 … S5 with removable tape and record the OD at each before the test: stations, directions and micrometer per SRM100-101 record R9.
10. Disassembly, never lever or knock: split clamp collar (100.0 bore) on the case OD at least 50 mm from the end being extracted (both ends), clear of the screw heads at z 13 (ϕ106.04 sweep, ref), the labels and the marking at z 20–45 (z from that end face); bridge and 2 × M6 draw screws in the closure's puller threads, 2 × M6×1-6H ↧8 / ⌀5 ↧10 at θ 10° and 190° (item 2 r 41, item 6 r 37.5). Item 17 comes out of item 2 as the insert does: holder on its forward face in an arbor press, push sleeve through the lip bore.
11. USED ON: SRM-100 motor (Helios) – proof before the first firing and after any repair.

HYDROSTATIC PROOF – PROCEDURE AND ACCEPTANCE

1. Medium: water. Proof pressure 10.5 MPa = 1.5 × MEOP (7.0 MPa).
2. Water-fill with the motor vertical, AFT END UP: item 17's 1/8-27 NPT port (view B) is the motor's only port (item 6 has none) and its cup floor the cavity's highest point. Items 2 and 17 make one flat ceiling at z 38, so tilting would drive bubbles out to the bore: fill from the bottom instead (SET-UP schematic). Pass a ϕ4 × 800 PE or nylon dip tube, end cut at 45°, down the port and the ϕ6 passage to rest on item 6's inboard face; fill slowly so the water rises level. Withdraw it, top up until the port overflows, fit the rig tee (bleed valve on top, pump line on the side branch), pump slowly with the bleed open until only water comes out, then close it. Trapped air is stored energy.
3. Set the pump's relief valve to 11 MPa before pressurising. Stand behind a barrier while the motor is under pressure.
4. Pressurise in steps 3.5 → 7.0 → 10.5 MPa, holding 300 s at each. Accept a hold if the pressure decays ≤ 2 % over it and there is no visible leak or weep.
5. Measure the case OD at the marked stations (S1 … S5, SRM100-101 record R9) before and after. Accept a permanent set ≤ 0.02 mm at every station.
6. Re-gauge every bolt hole against its baseline (SRM100-101 record R7). Retire the end if any hole's axial length has grown by more than 0.05 mm, or its ligament has shrunk by more than 0.05 mm, from that hole's own baseline.
7. After the first proof, take the hardness / conductivity baseline on each case end (SRM100-101 record R8): HRB or Webster B, and eddy-current %IACS, at θ 50°, 130°, 230°, 310° (between the holes); aft end z 5–25 and z 100, fwd end z 665.8–685.8 and z 590.8. 6061-T6 reference: about 95 HB (about 60 HRB), Webster B ≥ 15, 40–46 %IACS. Later readings on the same instrument and scale; after a firing, retire the end if > 46 %IACS or hardness more than 10 % below this baseline (SRM100-101 R10).
8. Re-proof after any repair, any retired-and-replaced part, or a temperature label reading ≥ 232 °C.
9. Record against the serials of the case / closure pair and of item 17: the step pressures and hold decays; the OD before and after, the hole gauging and the hardness baseline on the SRM100-101 records R9, R7 and R8.
10. Rig tee into item 17's 1/8-27 NPT port (ASME B1.20.1): Loctite 567 or equal on the male thread, first 2 threads clear; no PTFE tape. By the test operator, not defined in the design data: gauge range and accuracy, the isolation valve that holds the pump off during the holds, valves, fittings and hoses, water quality and temperature, drain and dry-out.
11. After the test: drain through item 17's port, motor aft end down (or siphon out through the dip tube), then strip (PROOF BUILD note 10) and dry. Item 2's bore and gland must be dry before the graphite insert goes in. The flight build takes new O-rings and screws (items 13–16); the full replace / re-use list for every firing is on SRM100-000 sheet 2.

PROOF SET-UP (SCHEMATIC, NOT TO SCALE)

TEST ORIENTATION: AFT END UP



Gauge, tee, bleed and isolation valves, fittings and hoses: by the test operator (PROCEDURE note 10).

FITS AND SEALS (ref, from the detail drawings)

JOINT	HOLE / BORE (item)	SHAFT / LAND (item)	FIT (ref)
Case bore / closure lands	seal zones ϕA +0.030/0 (1)	ϕA − 0.06 0/−0.02 (2, 6)	clearance 0.060–0.110
Case O-ring glands, 4	groove ϕA − 5.67 0/−0.04 × 4.80 +0.08/0 (2, 6)	AS568-237 (13)	squeeze 19.7 %
Holder bore / dummy OD	ϕ63.58 +0.025/0 (2)	ϕ63.50 0/−0.05 (17)	clearance 0.080–0.155
Insert gland	groove ϕ69.14 +0.05/0 × 4.80 +0.08/0 (2)	AS568-230 (14)	squeeze 20.1 %
Seat depth / dummy length	seat 83.21 +0.05/0 (2)	83.21 0/−0.05 (17)	0–0.10 below face
Screw holes, 2 × 18	ϕ6.40 +0.03/0 (1)	M6 tapped (2, 6)	match-drilled pair

A = the smallest reading in each seal zone of item 1, 92.000–92.070, zone A +0.030/0 (so ≤ 92.10), per zone: SRM100-101 R4. Items 2 and 6 are cut to their own end's A; at A = 92.000: land ϕ91.94, groove ϕ86.33.

PROOF SCHEDULE (medium: water)

STEP	PRESSURE	HOLD	ACCEPT THE HOLD IF
1	3.5 MPa	300 s	decay ≤ 2 % over the hold, no visible leak or weep
2	7.0 MPa = MEOP	300 s	decay ≤ 2 % over the hold, no visible leak or weep
3	10.5 MPa = 1.5 × MEOP	300 s	decay ≤ 2 % over the hold, no visible leak or weep

Relief valve set 11 MPa before pressurising. Proof 10.5 MPa = 1.5 × MEOP 7.0 MPa.

AFTER THE TEST

CHECK	ACCEPT / ACTION
Case OD at stations S1 … S5 (SRM100-101 R9), before / after	accept a permanent set ≤ 0.02 mm at every station
Every bolt hole (2 × 18), against its baseline (SRM100-101 R7)	retire the end if any hole's axial length has grown, or its ligament has shrunk, by more than 0.05 mm from that hole's own baseline
Case ends, after the first proof	hardness / conductivity baseline, SRM100-101 R8 (PROCEDURE note 7)

Mass 6089.8 g, water not included: sum of qty × unrounded mass each. Items 1, 2, 6, 13–16 from out/mass_budget.json; item 17 from its STEP, adaptive BRepGProp volume × 8.00 g/cm³. Tooling (hole plugs, push sleeve, puller bridge, clamp collar) is not listed.

ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL	MASS EACH g
17	1	SRM100-120	Proof dummy (replaces item 3)	304 stainless, ASTM A479	1414.1
16	18	ISO 7380-1 M6×16-010.9	Button-head screw, fwd	steel 010.9	4.7
15	18	ISO 7380-1 M6×10-010.9	Button-head screw, aft	steel 010.9	3.4
14	1	AS568-230	O-ring, insert seal	FKM 90A (Parker V0709-90 or equal)	3.8
13	4	AS568-237	O-ring, case seal	FKM 90A (Parker V0709-90 or equal)	5.1
6	1	SRM100-103	Forward closure	6061-T651	770.5
2	1	SRM100-102	Nozzle holder (aft closure)	304 stainless	1501.2
1	1	SRM100-101	Motor case	6061-T6 drawn tube	2234.4

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
See parts list	6089.8 g	ISO 2768-mK	–	–

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
ASSEMBLY DRAWING	NOT CHECKED – FOR REVIEW	1:2		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	HYDROSTATIC PROOF ASSEMBLY Flight case and closures, proof dummy SRM100-120 in the insert gland

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-001	A	1 / 1